1. Any square matrix can be multiplied by itself any (whole) number of times. So, when $\mathbf{A}$ is square, we can evaluate its *powers* $\mathbf{A}^2$, $\mathbf{A}^3$, etc. Given a matrix $\mathbf{A}$ below, compute $\mathbf{A}^k$ for $k = 2, 3, 4$, what will higher powers give?

   1.1. $\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

   1.2. $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$

2. Consider transformations given by $\mathbf{R}^3 \Rightarrow \mathbf{R}^3$ given by

   a) $S(x_1, x_2, x_3) = (x_2, x_3, 3x_1)$

   b) $T(x_1, x_2, x_3) = (2x_3, x_1 + x_2, 3x_1 - x_3)$

   Find the matrix that represents $T \circ S$.

3. Verify that these pairs of matrices are inverses.

   3.1. $\begin{bmatrix} 2 & 1 & 0 \\ 9 & 4 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} -4 & 1 & 0 \\ 9 & -2 & 0 \\ -9 & 2 & 1 \end{bmatrix}$

3.2. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$

4. Find the determinant of each matrix. What values of a (if any) make each matrix singular?

4.1. $\begin{bmatrix} 1 & 1 \\ a & a^2 \end{bmatrix}$

4.2. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & a & a^2 \end{bmatrix}$

4.3. $\begin{bmatrix} a & \sqrt{8} & 0 \\ \sqrt{8} & a & \sqrt{8} \\ 0 & \sqrt{8} & a \end{bmatrix}$